

Measures of skewness:

→ The lack of symmetry in the frequency distribution is known as skewness.

Types of skewness

① Symmetrical or No skewness:

→ A frequency distribution is called symmetrical if mean = median = mode.

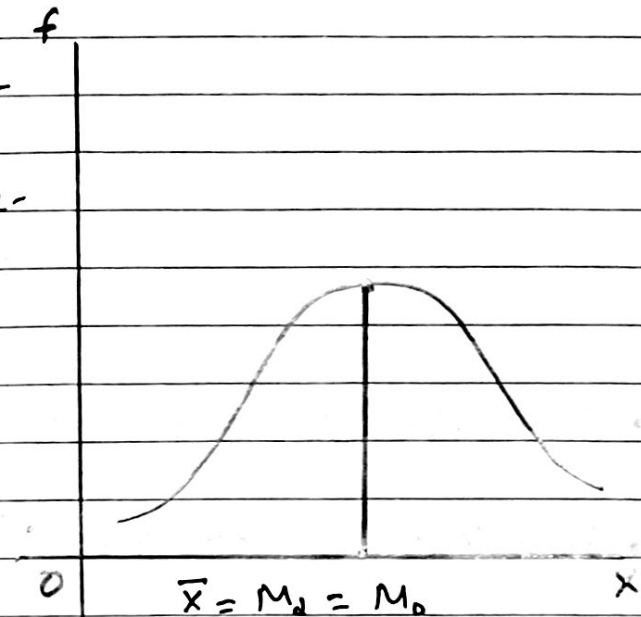


fig- Symmetrical distribution

② Positive or Right skewness.

→ A frequency distribution is called positive or Right skewed if mean > median > Mode

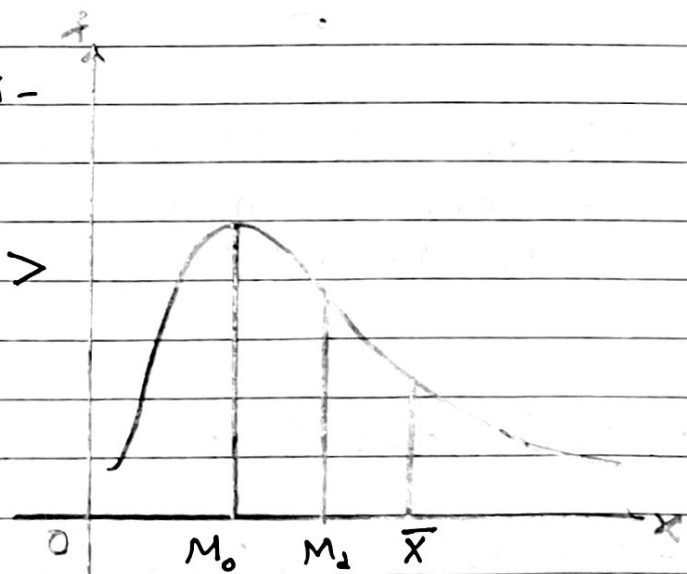


fig- positive or right skewed
Skewness

③ Negative or left skewness

→ A frequency distribution is called negative or left skewed if $\text{Mean} < \text{Median} < \text{Mode}$

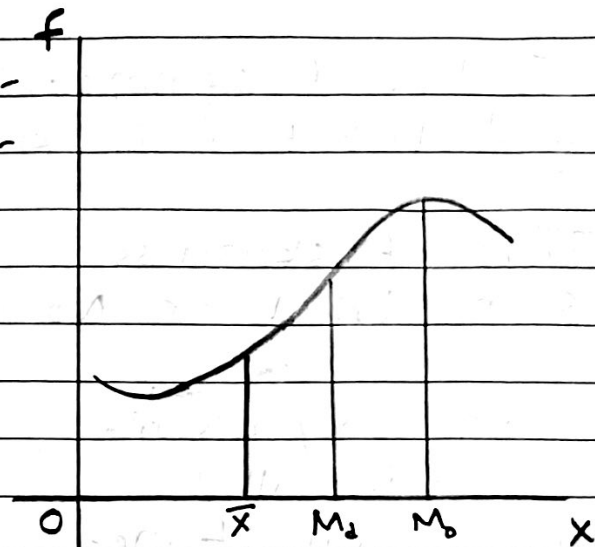


fig- Negative or left skewness

- Absolute measure of skewness:

→ $S_k = \bar{X} - M_o$ or, $S_k = \bar{X} - M_d$

- Relative measure of skewness:

Karl Pearson's coefficient of skewness $S_k(P)$

$$= \frac{\bar{X} - M_o}{\sigma} \quad [-1 \text{ to } 1]$$

If mode is ill defined, then we use the following formula,

$$S_k(P) = \frac{3(\bar{X} - M_d)}{\sigma} \quad [-3 \text{ to } 3]$$

Interpretation:

- If $S_k(P) = 0$, then there is no skewness

(symmetrical).

- If $S_k(P) \geq 0$, then the distribution is right or positively skewed.
- If $S_k(P) \leq 0$, then the distribution is left or negatively skewed.

Exercise 15.2

3. By calculating the Karl Pearson's coefficient of skewness test the symmetry of the following data.

a) 43, 12, 31, 20, 17, 26, 35, 49, 5, 37

→ Soln,

Arranging the given data in ascending order

5, 12, 17, 20, 26, 31, 35, 37, ~~43~~ 43, 49

For median:

No. of obs = $n = 10$

$$\therefore \text{Median}(M_e) = \left(\frac{n+1}{2} \right)^{\text{th}} \text{ item}$$

$$= \left(\frac{10+1}{2} \right)^{\text{th}} \text{ item}$$

$$= \left(\frac{11}{2} \right)^{\text{th}} \text{ item}$$

$$= \frac{26+31}{2}$$

$$= 28.5$$

X	X ²
5	25
12	144
17	289
20	400
26	676
31	961
35	1225
37	1369
43	1849
49	2401
$\Sigma x = 275$	$\Sigma x^2 = 9339$

For mean,

$$\bar{X} = \frac{\Sigma x}{n}$$

$$= \frac{275}{10}$$

$$= 27.5$$

And,

$$\sigma = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

$$= \sqrt{\frac{9339}{10} - (27.5)^2}$$

$$= \sqrt{933.9 - 756.25}$$

$$= 13.32$$

Finally,

$$S_k(p) = \frac{3(\bar{X} - M_d)}{\sigma}$$

$$= \frac{3(27.5 - 28.5)}{13.32}$$

$$= -0.225$$

$\therefore$ The given distribution is negatively skewed.

4. For the frequency distribution given below, Calculate the coefficient of skewness based on mean, mode and standard deviation.

Also comment on the symmetry of the data.

→ Soln,

$$\text{Correction factor} = \frac{\text{Lower limit of 2nd class} - \text{Upper limit of 1st class}}{2}$$

$$= \frac{20 - 19}{2}$$

$$= 0.5$$

Now,

We correct the class intervals by lower limit - correction factor and Upper limit + correction factor.

class	f	Mid value (x)	fx	fx ²
9.5-19.5	5	14.5	72.5	1051.25
19.5-29.5	9	24.5	220.5	5402.25
29.5-39.5	13	34.5	448.5	15473.25
39.5-49.5	20	44.5	890	39605
49.5-59.5	15	54.5	817.5	44553.75
59.5-69.5	8	64.5	516	33282
N=70			$\Sigma fx = 2965$	$\Sigma fx^2 = 139367.5$

Now,

$$\bar{X} = \frac{\Sigma fx}{N} = \frac{2965}{70}$$

$$= 42.35$$

Mode:

Modal class = (39.5 - 49.5) as it has the highest frequency 20.

So,

$$L = 39.5, f_0 = 13, f_1 = 20, f_2 = 15, i = 10$$

$$\begin{aligned} \therefore M_0 &= L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times i \\ &= 39.5 + \frac{20 - 13}{2 \times 20 - 13 - 15} \times 10 \\ &= 39.5 + \frac{7}{12} \times 10 \\ &= 39.5 + 5.83 \\ &= 45.33 \end{aligned}$$

For S.D

$$\begin{aligned} \sigma &= \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} \\ &= \sqrt{\frac{139367.5}{70} - \left(\frac{2965}{70}\right)^2} \\ &= 14.02 \end{aligned}$$

Finally,

$$\begin{aligned} S_k(P) &= \frac{\bar{X} - M_0}{\sigma} \\ &= \frac{42.35 - 45.33}{14.02} \\ &= -0.2120 \end{aligned}$$

$\therefore$ The distribution is negatively skewed.

- ⑥ Calculate Karl Pearson's coefficient of skewness from the following data.
Also, comment on the symmetry of the data.

→ Soln,

Marks	f	Mid value (X)	fx	fx ²	Cf
0-10	10	5	50	250	10
10-20	40	15	600	9000	50
20-30	20	25	500	12500	70
30-40	0	35	0	0	70
40-50	10	45	450	20250	80
50-60	40	55	2200	121000	120
60-70	16	65	1040	67600	136
70-80	14	75	1050	78750	150
Σf N = 150 Σfx = 5890 Σfx ² = 309350					

For median,

$$\text{Location} = \left(\frac{N}{2}\right)^{\text{th}} \text{ term} = \left(\frac{150}{2}\right)^{\text{th}} \text{ term} \\ = 75^{\text{th}} \text{ term}$$

The C.F just greater than 75 is 80. So,
M_d Class = 40-50

$$M_d = L + \frac{N/2 - cf}{f} \times i \\ = 40 + \frac{75 - 70}{10} \times 10 \\ = 40 + 5 \\ = 45$$

For mean,

$$\begin{aligned}\bar{X} &= \frac{\sum fx}{N} \\ &= \frac{5890}{150} \\ &= 39.26\end{aligned}$$

For standard Deviation,

$$\begin{aligned}\sigma &= \sqrt{\frac{\sum fx^2}{N} - \left(\frac{\sum fx}{N}\right)^2} \\ &= \sqrt{\frac{309350}{150} - (39.26)^2} \\ &= \sqrt{2062.33 - 1541.8711} \\ &= \sqrt{520.46} \\ &= 22.81\end{aligned}$$

Now,

$$\begin{aligned}S_k(P) &= \frac{3(\bar{X} - M_2)}{S.D(\sigma)} \\ &= \frac{3(39.26 - 45)}{22.81} \\ &= -0.754\end{aligned}$$

$\therefore$ The given distribution is negatively skewed.

(i)

- a) For a group of 10 items, $\sum x = 452$, $\sum x^2 = 24270$ and mode = 43.7, find the pearsonian coefficient of skewness.

$\hookrightarrow$ Soln,

$$N = 10$$

$$\sum x = 452$$

$$\sum x^2 = 24270$$

$$M_0 = 43.7$$

Now,

$$S_k(P) = \frac{\bar{X} - M_0}{\sigma}$$

For, mean,

$$\bar{X} = \frac{\sum X}{N} = \frac{452}{10} = 45.2$$

For standard deviation,

$$\begin{aligned} \sigma &= \sqrt{\frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2} \\ &= \sqrt{\frac{24270}{10} - (45.2)^2} \\ &= \sqrt{2427 - 2043.04} \\ &= \sqrt{383.96} \\ &\approx 19.59 \end{aligned}$$

Now,

$$\begin{aligned} S_k(P) &= \frac{45.2 - 43.7}{19.59} \\ &= \frac{1.5}{19.59} \\ &\approx 0.077 \end{aligned}$$

$$\therefore S_k(P) = 0.077$$

b) For a group of 20 items, if $\sum x = 120$, $\sum x^2 = 945$ and $M_0 = 7.5$, find the coefficient of skewness.

↳ Soln,

$$N = 20$$

$$\sum x = 120$$

$$\sum x^2 = 945$$

$$M_2 = 7.5$$

Now,

$$\bar{X} = \frac{\sum x}{N} = \frac{126}{21} = 6$$

$$\begin{aligned} \sigma &= \sqrt{\frac{\sum x^2}{N} - \left(\frac{\sum x}{N}\right)^2} \\ &= \sqrt{\frac{945}{21} - 6^2} \\ &= \sqrt{45 - 36} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

Finally,

$$\begin{aligned} S_k(P) &= \frac{3(\bar{X} - M_2)}{\sigma} \\ &= \frac{3(6 - 7.5)}{3} \\ &= -1.5 \end{aligned}$$

$$\therefore S_k(P) = -1.5$$

2.

a) In a certain distribution, the mean, median and coefficient of skewness are 45, 48 and 0.4 respectively. Find standard deviation and the coefficient of variable.

→ soln,

$$\bar{X} = 45$$

$$M_2 = 48$$

$$S_k(P) = 0.4$$

$$S.D (\sigma) = ?$$

We know that,

$$S_k(P) = \frac{3(\bar{X} - M_o)}{\sigma}$$

$$0.4 = \frac{3(45 - 48)}{\sigma}$$

$$\text{or, } \sigma = \frac{3 \times -3}{0.4}$$

$$\text{or, } \sigma = \frac{-9}{0.4}$$

$$\text{or, } \sigma = -22.5$$

$\therefore$ standard deviation = 22.5

Now,

$$\text{Coe. of Variation} = \frac{\sigma}{\bar{X}} \times 100\%$$

$$= \frac{22.5}{45} \times 100\%$$

$$= 50\%$$

b. A frequency distribution gives the following results.

Mean = 30, SD = 8. Karl Pearson's coefficient of skewness = 0.40.

Find the median and mode of distribution.

→ Soln,

$$S_k(P) = \frac{\bar{X} - M_o}{\sigma}$$

$$\Rightarrow 0.4 = \frac{30 - M_o}{8}$$

$$\Rightarrow 3.2 = 30 - M_o$$

$$\Rightarrow M_o = 30 - 3.2$$

$$\therefore M_o = 26.8$$

Again,

$$M_0 = 3M_2 - 2\bar{X}$$

$$26.8 = 3M_2 - 2 \times 30$$

$$\text{or, } 3M_2 = 26.8 + 60$$

$$\text{or, } M_2 = \frac{86.8}{3}$$

$$\therefore M_2 = 28.93$$

3.

b)	X	f	fx	fx ²
	100	5	500	50000
	110	6	660	72600
	120	8	960	115200
	130	6	780	101400
	140	4	560	78400
	N = 29		Σfx = 3460	Σfx^2 = 417600

$$\bar{X} = \frac{\Sigma fx}{N}$$

$$= \frac{3460}{29} = 119.31$$

$$\begin{aligned} \text{S.D (s)} &= \sqrt{\frac{\Sigma fx^2}{N} - \left(\frac{\Sigma fx}{N}\right)^2} \\ &= \sqrt{\frac{417600}{29} - (119.31)^2} \\ &= \sqrt{14400 - 14234.95} \\ &= \sqrt{165.05} \\ &= 12.84 \end{aligned}$$

Since the highest frequency is 8 so
mode = 120

Now,

$$\begin{aligned} S_k(P) &= \frac{\bar{X} - M_0}{\sigma} \\ &= \frac{119.31 - 120}{12.84} \\ &= \frac{-0.69}{12.84} \\ &= -0.054 \end{aligned}$$

∴ The given distribution is negatively skewed.

C)	X	f	Mid value (m)	fm	fm ²	Cf
	0-10	15	5	75	375	15
	10-20	30	15	450	6750	45
	20-30	25	25	625	15625	70
	30-40	20	35	700	24500	90
	40-50	10	45	450	20250	100
	N=100		$\Sigma fm = 2300$		$\Sigma fm^2 = 67500$	

Now,

$$\bar{X} = \frac{\Sigma fm}{N} = \frac{2300}{100} = 23$$

$$\begin{aligned}
 S.D(\sigma) &= \sqrt{\frac{\Sigma fm^2}{N} - \left(\frac{\Sigma fm}{N}\right)^2} \\
 &= \sqrt{\frac{67500}{100} - (23)^2} \\
 &= \sqrt{675 - 529} \\
 &= \sqrt{146} \\
 &= 12.08
 \end{aligned}$$

For median,

$$\begin{aligned}
 \text{Location} &= \frac{N}{2} \text{ term} = \left(\frac{100}{2}\right) \text{ term} \\
 &= 50^{\text{th}} \text{ term}
 \end{aligned}$$

The C.F is ~~great~~ lies on (20-30) class

$$\begin{aligned}
 M_d &= L + \frac{N/2 - cf}{f} \times i \\
 &= 20 + \frac{50 - 45}{25} \times 10
 \end{aligned}$$

$$= 20 + \frac{50}{25}$$

$$= 20 + 2$$

$$= 22$$

Again,

$$S_k(P) = \frac{3(\bar{x} - M_d)}{\sigma}$$

$$= \frac{3(23 - 22)}{12.08}$$

$$= \frac{3}{12.08}$$

$$= 0.24$$

∴ The given distribution is positively skewed.

Q5 Calculate the Pearson's coefficient of skewness of ages of 40 persons from the following data.

Also, comment on the symmetry of the data.

X	f	Mid value 1st (M)	fM	fM ²	cf
0-10	0	5	0	0	0
10-20	5	15	75	1125	5
20-30	10	25	250	6250	15
30-40	15	35	525	18375	30
40-50	8	45	360	16200	38
50-60	2	55	110	6050	40
N=40			ΣfM=1320	ΣfM ² =48000	

Now,

$$\bar{X} = \frac{\sum fM}{N}$$

$$= \frac{1320}{40}$$

$$= 33$$

$$\therefore \bar{X} = 33$$

$$\begin{aligned} S.D = \sigma &= \sqrt{\frac{\sum fM^2}{N} - \left(\frac{\sum fM}{N}\right)^2} \\ &= \sqrt{\frac{48000}{40} - (33)^2} \\ &= \sqrt{1200 - 1089} \\ &= \sqrt{111} \\ &= 10.53 \end{aligned}$$

For median,

$$\text{location} = \left(\frac{N}{2}\right)^{\text{th}} \text{ term}$$

$$= \left(\frac{40}{2}\right)^{\text{th}} \text{ term}$$

$$= 20^{\text{th}} \text{ term}$$

 $\therefore M_d$ lies on (30-40) class

$$M_d = L + \frac{N/2 - cf}{f} \times i$$

$$= 30 + \frac{20 - 15}{15} \times 10$$

$$= 30 + \frac{50}{15}$$

$$= 30 + 3.33$$

$$= 33.33$$

Finally,

$$S_k(P) = \frac{3(\bar{x} - M_2)}{\sigma}$$

$$= \frac{3(33 - 33.33)}{10.53}$$

$$= \frac{3 \times -0.33}{10.53}$$

$$= \frac{-1}{10.53}$$

$$= -0.09$$

∴ The distribution is negatively Skewed.